Time-travel versioning in Delta

The freedom and flexibility of open file formats make it possible to integrate with new and existing external tooling. Delta Lake, in particular, offers unique support to create data products thanks to features such as time-travel versioning, exceptional speed, and the ability to update and merge changes. Time travel, in this context, refers to the capability of querying different versions of your data table, allowing you to revisit the state of the table before your most recent changes or transformations (see *Figure 1.5*). The more obvious use is to back up after making a mistake rather than writing out multiple copies of the table as a safety measure. A possibly less obvious use for time travel is reproducible research. You can access the data your model was trained on in the previous week without creating an additional copy of the data. Throughout the book, we will detail features of the Data Intelligence Platform you can use to facilitate reproducible research. The following figure shows you how the previous version of a table, relative to a timestamp or a version number, can be queried.

```
-- view table as it was by timestamp
SELECT * FROM ml_in_action.favorita_forecasting.favorita_transactions TIMESTAMP AS OF '2023-09-29 22:57:25';

-- view table as it was by version
SELECT * FROM lakehouse_in_action.favorita_forecasting.favorita_transactions VERSION AS OF 1;
```

Figure 1.5 – A code example of the querying techniques to view previous versions of a table

The speed of Databricks' optimized combination

Next, let us discuss the speed of Databricks' lakehouse architecture. In November 2021, Databricks set a new world record for the gold standard performance benchmark for data warehousing. The Barcelona Computing Group shared their research supporting this finding. This record-breaking speed resulted from the Databricks' engines (Spark and Photon) paired with Delta (see the *Databricks Sets Official Data Warehousing Performance Record* link in the *Further reading* section).

The additional benefits of Delta

Delta's impressive features include **change data feed** (**CDF**), **change data capture** (**CDC**), and **schema evolution**. Each plays a specific role in data transformation in support of ML.

Starting with Delta's CDF capability, it is exactly what it sounds like – a feed of the changed data. Let's say you have a model looking for fraud, and that model needs to know how many transaction requests have occurred in the last 10 minutes. It is not feasible to rewrite the entire table each time a value for an account needs to be updated. The feature value, or in this case, the number of transactions that occurred in the last 10 minutes, needs to be updated only when the value has changed. The use of CDF in this example enables updates to be passed to an **online feature store**; see *Chapters 5* and *6* for more details.